

- 36** Carbon dioxide is continually produced by respiring cells. It diffuses into red blood cells and dissociates into hydrogen ions (H^+) and hydrogen carbonate ions (HCO_3^-). HCO_3^- then diffuses out of the cell in exchange for chloride ions (Cl^-) in the chloride shift.

What is the importance of the chloride shift?

- A** It helps to maintain a neutral pH in the red blood cell by acting as a buffer.
- B** It maintains a balance of positive and negative charge between the cell and the plasma.
- C** It causes the oxygen dissociation curve to shift to the left, releasing more oxygen.
- D** It prevents carbon dioxide combining with haemoglobin, allowing haemoglobin to combine with oxygen.